

Thong Nguyen

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EDUCATION

University of California, Berkeley

B.S. Electrical and Computer Engineering

GPA: 3.15

Expected Dec 2027

SKILLS

Hardware and EE tools: KiCad, LTSpice, Multimeter, Power Supply, Oscilloscope, Signal Generator

Courseworks: The Structure and Interpretation of Computer Programs, Great Ideas of Computer Architecture, Intro to Signals, Introduction to Circuits, Hands-On PCB Engineering, Microelectronic Devices, Computer Architecture and Engineering*, Integrated-Circuit Devices*, Linear Integrated Circuits*

WORK EXPERIENCE

Electronics Sustaining Engineer Intern

May 2026 - August 2026

Masimo Corporation, Irvine, CA

- Assisted in developing the calibration procedure for a tester board and identified the root cause of signal distortion during clock signal pin calibration.
- Analyzed board schematics to interpret and verify expected calibration results.
- Programmed SPI communication on a Nordic development kit to analyze failed EEPROMs caused by data corruption, impacting approximately 3% of production.

Volunteer Research Assistant

October 2024 - May 2025

Hero Lab at University of California Irvine, Irvine, CA

- Assisted a PhD student in developing a wearable watch using ECG and SPG signals to measure heart rate and blood pressure.
- Supported the development of a custom Raspberry Pi 5 with object detections features and AI Camera.

Software Developer Intern

July 2024 - August 2024

Magnima LLC, Midway City, CA

- Updated embedded C firmware using the Nordic nRF5 SDK for the nRF52832 SoC for maintenance.
- Flashed, debugged, and brought up custom PCBs via USB-UART, resolving driver, and hardware–software interface issues.

PROJECTS

Treble Boost Audio Effects Circuit

- Designed an op-amp treble boost audio circuit powered by a ± 15 V supply, and simulated transient and AC responses in LTSpice to verify $\geq 10\times$ gain and frequency response.
- Extracted low- and high-cutoff frequencies from -3 dB points and generated Bode magnitude and phase plots in both simulation and experiment.
- Breadboarded and validated the circuit using a Digilent AD2, applying a guitar-modeled sinusoidal input and confirming distortion through amplified audio output.

Pitch Detector

- Designed a USB-C–powered Li-ion power management system in KiCad using BQ24074 and TPS62162, enabling simultaneous battery charging and ESP32 operation.
- Implemented 5 V USB-C input, 4.2 V Li-ion charging, and a regulated 3.3 V rail, and programmed the ESP32 (Arduino) to detect and display musical notes.
- Completed DFM-compliant PCB layout, setting 6 mil trace widths, 24/12 mil vias, and correctly sizing thermal and mechanical features for fabrication.